



**University of Vocational Technology
Faculty of Information Technology
Department of Software Technology**

**Bachelor of Technology
In
Software Technology**

**Final Year Project
Final Project Proposal**

Construction Equipment and Machinery Registration System

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Construction Equipment and Machinery Registration System

1. Introduction

The construction industry relies heavily on machinery and equipment for efficient operations. However, the absence of a centralized system to register, monitor, and manage these assets often leads to underutilization, poor maintenance, and administrative difficulties. To address these challenges, this project proposes a Construction Equipment and Machinery Registration System, a digital solution that enables the registration and monitoring of heavy construction equipment through a centralized online platform integrated with the Construction Industry Development Authority (CIDA).

This system will serve construction contractors, property developers, machinery owners, and investors by offering reliable information about available machinery and equipment across various locations. It will also promote transparency, ensure compliance, and optimize equipment utilization throughout the construction industry.

2. Background

Currently, machinery owners manually manage their assets using paperwork or isolated records. This leads to inefficient recordkeeping, lack of visibility, and difficulty identifying ownership or operational status. The Construction Industry Development Authority (CIDA) mandates machinery registration under national regulations, but this process remains largely offline.

By introducing a digital registration and system, the registration process can be automated, machinery data can be verified electronically. This system will assist both government authorities and private stakeholders in maintaining a complete, updated inventory of active construction machinery.

3. Problem in Brief

Many challenges exist in the current manual process:

- Lack of centralized machinery data.
- Difficulty locating available equipment near specific project sites.
- Poor communication between machinery owners and contractors.
- Limited transparency in ownership verification and maintenance records.
- Increased equipment idle time due to inefficient coordination.

Therefore, an integrated online registration system is essential to enhance accessibility, operational efficiency, and accountability in machinery management.

4. Aim and Objectives

✓ **Aim:**

To design and develop a centralized system for the registration and monitoring of construction machinery and equipment registered under the Construction Industry Development Authority (CIDA).

✓ **Objectives:**

- To allow machinery owners to register and update their equipment details online.
- To enable CIDA officers to verify and approve or reject equipment registration requests.
- To provide contractors with an efficient search feature to find available equipment near specific project sites.
- To generate analytical reports on machine usage, condition, and registration validity for administrative purposes.
- To ensure proper compliance with the *Construction Industry Development (Registration of Heavy Construction Machinery and Equipment) Rules, 2019*.

5. Benefits for the Construction Machinery and Equipment Owners

- **Increased Hiring Opportunities:**
Contractors, property developers, and investors can easily find and contact registered machinery owners through the platform, resulting in more business opportunities.
- **Better Equipment Visibility:**
Machinery and equipment registered under CIDA will be visible to all verified users, making it easier to rent or hire.
- **Reduced Idle Time:**
Idle equipment can be located by nearby projects, ensuring continuous use and revenue generation.
- **Trust and Credibility:**
Being listed as a registered owner under the CIDA system increases credibility and attracts more clients.

6. Scope Overview

This system is designed to digitalize and automate the registration and management process of heavy construction machinery and equipment. It will serve the Construction Industry Development Authority (CIDA), machinery/equipment owners, and construction contractors seeking to hire or verify equipment.

✓ **Inclusions (What the System Will Do)**

- i. **User Registration and Authentication**
 - Allow registration for machinery/equipment owners, CIDA officers, and contractors.
 - Secure login for all users.

ii. Machinery and Equipment Registration

- Enable owners to submit registration applications online (as required under Rule 2 and 3 of the Gazette).
- Allow document uploads such as:
 - Motor Traffic Registration Certificate
 - Revenue License (for road-running machinery)
 - Affidavit of ownership
 - Serviceability report from a qualified Mechanical Engineer.

iii. CIDA Verification and Approval

- CIDA officers can verify documents and approve or reject registration requests.
- Automated notification of approval, rejection, or renewal requirements.

iv. Certificate Management

- Generate and store digital Certificates of Registration.
- Manage certificate validity and renewal periods (as stated under Rules 5 and 7).

v. Equipment Register Database

- Maintain a centralized digital register (as per Rule 9(a)) containing all registered machinery and equipment.
- Support regular updates and public accessibility (Rule 9(b)).

vi. Search and Reporting

- Enable contractors and public users to search registered machinery by type, owner, or location.
- Generate analytical reports for CIDA on equipment statistics, registration renewals, and compliance.

vii. Renewal and Revocation Handling

- Support online renewal applications before expiry.

- Record and manage certificate revocations where applicable (Rule 6).

✓ **Exclusions (What the System Will Not Do)**

- Equipment maintenance, servicing, or repair tracking.
- Financial transactions such as equipment rental or sales.
- GPS-based real-time tracking or location monitoring of machinery.
- Handling of appeals submitted under Section 53 of the Construction Industry Development Act.
- Physical inspection or validation of machinery — the system only records reports issued by qualified engineers.
- Integration with third-party insurance or taxation systems.

7. Methodology

The project will follow a structured Software Development Life Cycle (SDLC) consisting of the following phases:

i. Planning and Requirement Analysis

- Identify stakeholder requirements (CIDA officers, machinery owners, contractors).
- Define functional and non-functional requirements.

ii. System Design

- Develop system architecture and database schema.
- Design user interfaces for registration, search, and tracking.

iii. Development

- Implement front-end and back-end modules.
- Integrate GPS/IoT-based tracking where applicable.

iv. Testing

- Conduct unit, integration, and system testing.
- Ensure data validation, accuracy, and system reliability.

v. Deployment

- Host the system on a secure cloud server.
- Link the platform with the CIDA registration database.

vi. Maintenance and Support

- Periodically update system modules.
- Address user feedback and performance enhancements.

8. Resource Requirements

- Software Tools: Java/Spring Boot, MySQL, HTML, CSS,Php
- Design Tools: Figma, Adobe XD.
- Human Resources: Developers, database administrators, system analysts.

9. System Requirements

a) Functional Requirements

i. User Registration and Login

- Allow owners of heavy construction machinery to create accounts.
- Enable CIDA officers and administrators to securely log in and manage system functions.

ii. Equipment Registration

- Owners can submit online applications to register machinery and equipment as per the Gazette rules.
- The form must capture:
 - Machine type and model (as per *First Schedule*).
 - Registration number and make.
 - Ownership details (with affidavit upload).
 - Condition/status report (within six months, issued by a certified mechanical engineer).
 - Supporting documents: Certificate of Registration (Motor Traffic), Revenue License, etc.

iii. Document Upload and Verification

- Upload PDF or scanned documents.
- Admin (CIDA) can view, verify, and approve or reject applications.
- Record and store reasons for rejection.

iv. Certificate Generation

- After approval, the system automatically generates a digital Certificate of Registration (valid for a specific period, as per rule 5).
- Include registration number, owner details, and expiry date.

v. Renewal Management

- Notify owners 30 days before the certificate expiry (Rule 7(1)).
- Allow online renewal and payment of renewal fees (as per *Second Schedule*).

vi. Search and Public Access

- Maintain a searchable, publicly accessible register (Rule 9(b)).
- Allow search by owner, equipment type, registration number, or location.

vii. Reporting and Analytics

- Generate administrative reports (total registrations, renewals, rejected applications, machine categories, etc.).
- Export data to Excel/PDF for government records.

viii. Appeal Handling

- Enable submission and tracking of appeals (Rule 8) for rejected registrations.

b) Non-Functional Requirements

i. Security and Data Privacy

- Secure login (SSL/TLS, encrypted passwords).
- Role-based access (Owner, CIDA Officer, Administrator).
- Secure document storage and audit trail.

ii. Availability and Reliability

- Ensure 99% system uptime.
- Daily automatic data backup and recovery plan.

iii. Usability and Responsive Design

- User-friendly web interface accessible from desktops, tablets, and smartphones.

- Support Sinhala, Tamil, and English languages.

iv. Performance

- System should handle concurrent user sessions efficiently.
- Search and report generation must complete within 3–5 seconds.

v. Maintainability

- Modular code architecture for easy updates.
- Comprehensive admin dashboard for system management.

vi. Compliance

- Fully compliant with the Construction Industry Development (Registration of Heavy Construction Machinery and Equipment) Rules, 2019.

10. Progression Timeline

Expected Date		21-Jun-25	6-Nov-25	6-Nov-25	6-Nov-25	7-Nov-25	1st week Dec 2026	3rd week Dec 2026	4th week Dec 2026	2 nd week- Feb 2026	2nd week- Mar 2026	2nd week- Feb 2026	3rd week- Mar 2026	2 nd Week April 2026
Activity	Project group information google sheet updating													
	Draft proposal submission													
	Final proposal submission													
	Proposal presentation sideshow submission													
	Proposal presentation (On site)													
	1 st Progress report submission													
	Mid presentation sideshow submission													
	Mid presentation													
	2 nd Progress report submission													
	Final report(draft) submission													
	Final presentation sideshow submission													
	Final presentation													
	Final submission													

11.Reference

How to apply - Machinery Registration

<https://share.google/ZpZNEA3f9AVajwDLR>

482) 2116-49_E.pdf <https://share.google/Ja46cx4vgNIj1cF9T>